

were completed, but also when instrument settings were revised, or new reagents were prepared, etc. The mathematical model is the same, except that the effect for series is denoted B_i instead of L_i :

$$Z_{ij} = X + B_i + E_{ij}$$

Though the most classic application of this modeling is described in ISO 5725 standards for an interlaboratory study, the same mathematical approach is fully valid when applied to measures done in-house and under intermediate precision conditions. In both cases, data are structured according to a single factor, giving the name one-way ANOVA. As stated, the total variance is decomposed into two variances, one corresponding to the factor effects and the other to the errors, respectively. With more complex data structures, the decomposition must consider this complexity.

The starting point of the decomposition is quite simple, as illustrated by the basic equation (3.3). The total mean $\bar{\bar{Z}}$ is computed with the whole dataset and conventionally named the “grand mean.”

Basic equation

$$Z_{ij} - \bar{\bar{Z}} = (Z_{ij} - \bar{Z}_i) + (\bar{Z}_i - \bar{\bar{Z}}) \quad (3.3)$$

i th laboratory mean

$$\bar{Z}_i = \frac{\sum_{j=1}^J Z_{ij}}{I} \quad (3.4)$$

Grand mean

$$\bar{\bar{Z}} = \frac{\sum_{i=1}^I \bar{Z}_i}{I} \quad (3.5)$$

Squared basic equation

$$(Z_{ij} - \bar{\bar{Z}})^2 = [(Z_{ij} - \bar{Z}_i) + (\bar{Z}_i - \bar{\bar{Z}})]^2$$

Developed squared basic equation

$$\sum_i \sum_j (Z_{ij} - \bar{\bar{Z}})^2 = \sum_i \sum_j (Z_{ij} - \bar{Z}_i)^2 + \sum_i (\bar{Z}_i - \bar{\bar{Z}})^2 2 \sum_i \sum_j (Z_{ij} - \bar{Z}_i) \times (\bar{Z}_i - \bar{\bar{Z}}) \quad (3.6)$$

The basic equation is a simple set of additions and subtractions of different deviations. It may be surprising to square this equation, but if we try to compute the mean value of each difference, it is easy to see that all sums are equal to 0. Moreover, it can be demonstrated that the double cross-product of Eq. (3.6) is always equal to 0.

$$2 \sum_i \sum_j (Z_{ij} - \bar{Z}_i) \times (\bar{Z}_i - \bar{\bar{Z}}) = 0$$

The final Eq. (3.7) consists of three sums of squared deviations that can be related to the elements of the model, namely the total sum of squares, the repeatability or residual sum of squares, and the interlaboratory sum of squares.

Simplification

$$\sum_i \sum_j (Z_{ij} - \bar{\bar{Z}})^2 = \sum_i \sum_j (Z_{ij} - \bar{Z}_i)^2 + \sum_i (\bar{Z}_i - \bar{\bar{Z}})^2 \quad (3.7)$$

Total sum of squares = Repeatability sum of squares +
Interlaboratory sum of squares

$$\begin{aligned} \text{Simplified notation for in-house validation } SS_t &= SS_W + SS_B \\ \text{Simplified notation for interlaboratory study } SS_t &= SS_r + SS_L \end{aligned} \quad (3.8)$$

Equation (3.7) can be written in an equivalent abridged notation, which will be used below. The following conventions are used:

- Subscript *W* means *Within*
- Subscript *B* means *Between*-series in the context of in-house validation, and *L* means *Between*-laboratories in the context of interlaboratory.
- The notation *SS* for sum of squares will be very useful when developing an Excel worksheet, as explained in Section 3.4.

Finally, SS_W matches with repeatability. In the rest of the text, both notations are applied, and the same algorithm is used to compute precision parameters, either in the context of interlaboratory or in-house validation. ANOVA is nothing more than the application of Pythagoras's Theorem, where the square of the hypotenuse of a right-angled triangle equals the sum of squares of the other two sides. In other words, the total sum of squares is equal to the sum of squares due to repeatability and the sum of squares due to factor effects.

Figure 3.3 proposes a geometric illustration of this decomposition. Finally, the total sum of the squares of the deviations from the grand mean is decomposed into a sum of squares of deviations due to the laboratory (or series) effect and a residual sum of squares due to the repeatability error. More complete notation is required to accomplish the full calculation. Let us note SS_i the sum of the squares of the deviations of laboratory *i* from its own mean $\bar{Z}_i$.

Sum of squares of laboratory *i*

$$SS_i = \sum_j (Z_{ij} - \bar{Z}_i)^2 \quad (3.9)$$

Within (series or laboratory) sum of squares or Repeatability sum of squares

$$SS_r = SS_W = \sum_i SS_i \quad (3.10)$$

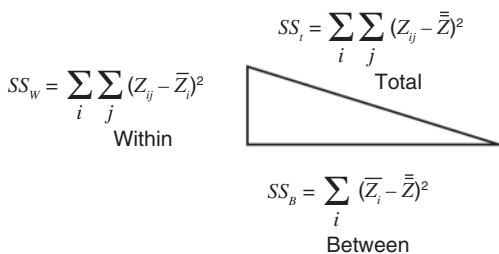


Figure 3.3 Geometric interpretation of the general ANOVA.

Total sum of squares

$$SS_t = \sum_i \sum_j (Z_{ij} - \bar{\bar{Z}})^2 \quad (3.11)$$

$$SS_t = SS_W + SS_B \quad (3.12)$$

Between-series sum of squares or Between-laboratories

$$SS_B = SS_L = SS_t - SS_W \quad (3.13)$$

The calculation by difference of the between-series (or -laboratories) sum of squares, denoted SS_B or SS_L is the simplest way to avoid any further calculation. It was proposed when modern computational means were not available. Therefore, in the first publication of ISO 5725 standards in 1994, this was presented as the standardized computation method.

There is a major downside when the effects of the factor (series or laboratories) are not significantly different, i.e. when the interlaboratory or between-series variance is close to 0. By calculation, the sum of squares and the corresponding variance may be negative, which is nonsense since, by construction, a variance must always be positive! As explained below, the recommendation is to set SS_L to 0. But other algorithms exist (Section 4.3.4).

Next, to calculate the variances, the SS are divided by the appropriate numbers of degrees of freedom. The degrees of freedom rely on an assumption about the distribution of residual errors E_{ij} . They are assumed to be all distributed according to identical Normal laws $\mathcal{N}(0, \sigma^2)$. This requirement is the same as for the residuals for the ordinary least-squares (OLS) method described in Section 2.3.1. If there is a way to circumvent this requirement by using weighted least-squares (WLS), there is not such a possibility with ANOVA. For instance, in Figure 3.2, most variances are identical and fulfill the basic assumption, except in Lab 03. Several statistical tests are proposed to verify this hypothesis in ISO 5725 standard. In the framework of in-house validation, we consider these tests to be useless.

The observed model in Figure 3.4a shows that the data from each laboratory are diversely scattered. This means they correspond to diverse distribution laws, each with a specific mean $\bar{Z}_i$ and standard deviation s_i . The size of the dotted circle illustrates the magnitude of the dispersion. When moving to the theoretical model developed by Fisher, it is assumed that all laboratories have the same standard deviation, corresponding to the repeatability standard deviation.

Figure 3.4b, illustrates the consequences of this assumption about variances in the modeling of data. In some way, the standard deviation of repeatability represents an average residual error standard deviation. At first glance, it may seem astonishing to enforce a single standard deviation and no longer consider the observed standard deviation of each laboratory. Behind this assumption stands the idea that the measuring procedure can be characterized by unique precision parameters, whoever applies it.

The differences between each laboratory mean $\bar{Z}_i$ and the reference value X assigned to the test sample are called the laboratory effects and are visually

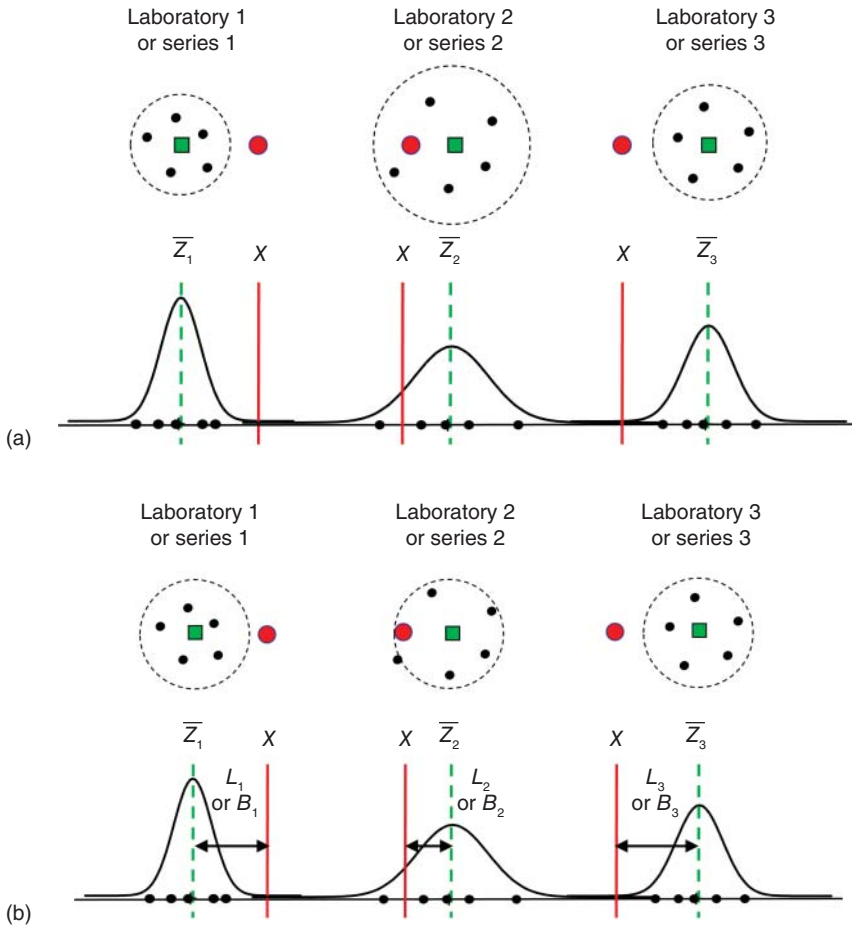


Figure 3.4 (a) ANOVA – observed model. (b) ANOVA – theoretical model.

illustrated by double horizontal arrows. They correspond to the random variables L_i (or B_i for the series) of the model. For one laboratory, it is essential to understand that the laboratory effect is not constant and will also randomly vary from one replicate to another.

Finally, the formulas applicable to obtain the precision variances and standard deviations are the following. They are presented in different equivalent notations. The same formulas are valid when data are obtained under reproducibility conditions or intermediate precision conditions, although trials are not done in the same experimental framework, several laboratories, or a single laboratory.

For clarity, the same notation I is used for different quantities (and names): for interlaboratory studies, it is the number of laboratories, while for in-house studies, it is the number of series.

Within-series variance or Repeatability variance

$$\begin{cases} s_r^2 = s_W^2 = \frac{SS_W}{N-I} \\ s_r^2 = \frac{SS_W}{I(J-1)} \end{cases} \quad (3.14)$$

$$s_r^2 = \frac{\sum_{i=1}^I \sum_{j=1}^J (Z_{ij} - \bar{Z}_i)^2}{I(J-1)} \quad (3.15)$$

Between-series variance or Between-laboratories variance (equivalent notations)

$$\begin{cases} s_L^2 = s_B^2 = \frac{\left(\frac{SS_B}{I-1} - s_r^2\right)}{J} \\ s_B^2 = \frac{(s_r^2 - s_L^2)}{J} \\ s_B^2 = \frac{\frac{\sum_{i=1}^I (\bar{Z}_i - \bar{Z})^2}{I-1} - s_r^2}{J} \end{cases} \quad (3.16)$$

Reproducibility variance

$$s_R^2 = s_r^2 + s_L^2 \quad (3.17)$$

Intermediate precision variance

$$s_{IP}^2 = s_r^2 + s_B^2 \quad (3.18)$$

Reproducibility standard deviation

$$s_R = \sqrt{s_r^2 + s_L^2} \quad (3.19)$$

Intermediate precision standard deviation

$$s_{IP} = \sqrt{s_r^2 + s_B^2} \quad (3.20)$$

3.2.2 Additional Parameters

3.2.2.1 Relative Standard Deviation of Parameters

A long-established parameter used for precision notation, proposed in many regulatory documents and analytical standards, consists in computing the relative standard deviation (*RSD*). This parameter, sometimes called the “coefficient of variation” or *CV*, is easily obtained as the ratio of a standard deviation and the corresponding average value. For instance, for one laboratory, the *RSD* is

$$RSD_i = \frac{s_i}{\bar{Z}_i} \times 100$$

This is a dimensionless parameter, generally expressed as a percentage. When estimating the precision, different *RSD* can be computed using the corresponding standard deviation, e.g. repeatability or intermediate precision, and the grand mean. According to our notations, the *RSD* of precision may be expressed at least in three different forms.

RSD of repeatability

$$RSD_r = \frac{s_r}{\bar{Z}} \times 100$$